

AI-based triage and decision support in mammography and digital tomosynthesis for breast cancer screening: a paired, noninferiority trial

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Artificial intelligence (AI) systems have been demonstrated to improve the accuracy of screening mammograms. Here this prospective, paired, noninferiority clinical trial evaluated whether AI could safely reduce workload by excluding low-risk exams from radiologist reading. Between March 2022 and January 2024, 31,301 women were included in the trial and underwent routine mammograms. Two reading strategies were applied in parallel: standard double-blind reading and partially autonomous AI-supported screening, where cases classified by AI as low risk were assessed as normal and the rest were double read with AI support. The primary outcomes were radiologist workload, cancer detection rate and recall rate. In the AI strategy, radiologist workload was 63.6% lower; the cancer detection rate was 15.2% higher (95% confidence interval 6.6%, 24.4%), increasing from 6.3 of 1,000 to 7.3 of 1,000, $P < 0.001$; and the recall rate was not noninferior and was 14.8% higher (95% confidence interval 9.0%, 20.6%). Subanalyses by modality highlighted a similar workload reduction in digital mammography (−62.1%) and digital breast tomosynthesis (−65.5%). However, in digital mammography, the cancer detection rate increased by 1.6 of 1,000 and the recall rate by 1.3%, while both remained stable in digital breast tomosynthesis. These results demonstrate the feasibility of a partially automated AI workflow in breast cancer screening, avoiding human reading of studies classified as low risk. ClinicalTrials.gov: [NCT04849776](https://clinicaltrials.gov/ct2/show/study/NCT04849776).

Breast cancer screening programs with digital mammography (DM) have been established for decades, with proven benefits in reducing breast cancer-related mortality¹. Still, screening programs face clinical challenges in terms of both missed cancers and false positive recalls². These can be improved by the use of digital breast tomosynthesis (DBT) and/or double reading, but at the cost of an ever-increasing reading workload³. In light of population aging, recommendations for starting screening at younger ages and the increasing shortage of qualified radiologists in some areas, alternative approaches using

artificial intelligence (AI) have been proposed to maintain or even improve screening quality without increasing radiologists' workload⁴.

Over the last few years, several retrospective studies have shown that radiologists improved their cancer detection accuracy when using an AI system as concurrent reading support^{5–8}. Furthermore, AI systems implemented as a stand-alone solution have been found to achieve human-like performance in simulated retrospective studies^{9–12}, classifying exams according to cancer probability and resulting in simulations of AI-based breast cancer strategies that suggested improved

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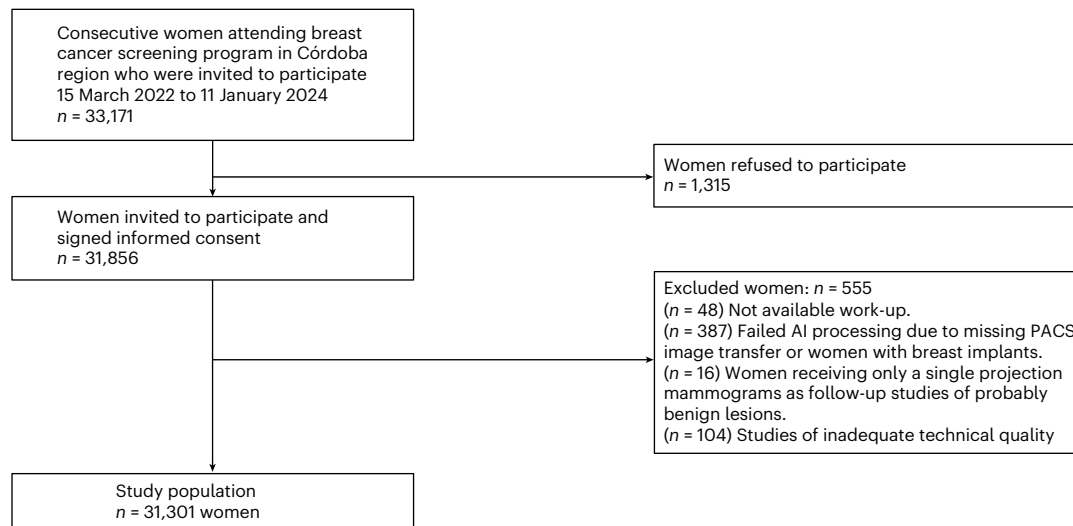


Fig. 1 | Study profile. Flowchart describing the inclusion of the study population, the women excluded and the different reasons for exclusion.

clinical outcomes^{4,13–17}. Several prospective studies have been designed to overcome the limitations of retrospective studies and to address and quantify the real-life impact of AI in breast cancer screening in which radiologists interact with AI. The first results from these studies have recently been published^{18–22}. Some of these studies showed that, compared to standard double reading, AI-supported double reading²¹ or using AI as a third reader²² can improve mammography screening metrics. Other prospective studies^{18–20} demonstrate that the use of AI to replace a reader in double-reading screening with DM is safe and effective, leading to a significantly reduced workload. A common trend in the outcomes of many retrospective and prospective studies is the high performance accuracy of AI in identifying large numbers of low-risk breast screening exams with a very high negative predictive value. As a consequence, the radiologist's reading could be completely avoided for such low-AI-risk exams, thus effectively introducing autonomous AI reading and partially automated screening. This has been simulated retrospectively in two large-scale studies with DBT population screening, demonstrating a negative predictive value of AI of 99.91% (ref. 17) and 99.97% (ref. 4) for the screening exams classified by AI as lowest risk.

The Artificial Intelligence in Breast Cancer Screening Program in Córdoba (AITIC) clinical trial is a prospective, paired, noninferiority, accuracy study (NCT04849776) that aims to confirm prospectively the results obtained in these studies by evaluating whether an AI system can be used to omit human reading completely in a large proportion of the screening exams classified as low risk for cancer, with noninferior results in the cancer detection rate (CDR) and recall rate (RR) compared to the standard strategy. This trial has a paired design, where each mammogram (either DM or DBT, depending on resource availability) has two reading strategies: the standard of care (double reading without AI support) and the AI-based intervention strategy (double reading with AI support if AI risk is higher than a predetermined threshold; alternatively, the exam will be automatically considered as normal).

This paired, noninferiority trial prospectively evaluates the use of an AI system in a real-world population-based breast cancer screening program for the autonomous and automated reading of a large number of examinations without requiring radiologist involvement in terms of workload, CDR and RR. This study includes DBT, which is increasingly used and recommended in screening programs²³.

Results

Women included in the trial and data characteristics

From 15 March 2022 to 11 January 2024, 33,171 consecutive women participated in the breast cancer screening program. On arrival at the screening site, they were informed of the study and offered the

opportunity to participate on a voluntary basis. A total of 31,856 women agreed to participate in this trial (AITIC, ClinicalTrials.gov ID NCT04949776) and signed the informed consent form (the patient information sheet and informed consent form are included in Zenodo²⁴). Of these, 555 women (1.7%) were excluded because of the different reasons explained in Fig. 1, resulting in 31,301 women being included (17,333 DM and 13,968 DBT).

Table 1 shows the characteristics of the participants included in the study and data recorded. The median age was 59 years old (interquartile range 64 to 54 years). This was the first screening for 11.3% of the participants. In terms of breast density, the distribution was as follows: A: 20.6%; B: 46.5%; C: 27.6%; and D: 5.3%. There were 252 screen-detected cancers between both strategies.

Primary outcomes

Table 2 shows the results obtained for the main outcomes (workload, CDR and RR) for the entire sample. The calculation is based on the patients who were included in the study. Extended Data Figs. 1 and 2 show the noninferiority plots for CDR and RR.

The standard strategy resulted in 62,602 radiologists screening readings and 198 screen-detected cancers. The CDR and RR were 6.3 of 1,000 (95% confidence interval (CI) 5.5 of 1,000, 7.2 of 1,000) and 4.8% (95% CI 4.6%, 5.0%), respectively. The AI strategy involved reading only 36.4% (11,384) of all screening exams, which resulted in 22,768 radiologist readings and 228 screen-detected cancers. The remaining studies (19,917 women with low-AI-risk exams) were classified as normal in this strategy without human reading. The CDR and RR were 7.3 of 1,000 (95% CI 6.3 of 1,000, 8.2 of 1,000) and 5.5% (95% CI 5.3%, 5.8%), respectively. For more details, see Table 2.

When comparing the results between the standard strategy and the AI strategy, there was a reduction in the workload of –63.6% (95% CI –64.2, –63.1 (–39,834 readings)); an increase in the CDR of 15.2% (95% CI 6.6%, 24.4%), which is an absolute difference of 1.0 of 1,000 (95% CI 0.4 of 1,000, 1.5 of 1,000; $P < 0.001$); and an increase in the RR of 14.8% (95% CI 9.0%, 20.6%), which is an absolute difference of 0.7% (95% CI 0.4%, 1.0%). Therefore, the CDR in the AI strategy was noninferior and statistically superior to that of the standard strategy while RR was not noninferior to the standard strategy. See Table 2 for more details.

Secondary outcomes and screening performance outcomes by modality

The positive predictive value (PPV) was similar in both strategies: 13.19% (95% CI 11.50%, 14.90%) in the standard strategy and 13.23% (95% CI 11.63%, 14.83%) in the AI strategy (absolute difference of 0.04% (95% CI 2.3%, 2.4%)).

Table 1 | Characteristics of the patients included in the study and data recorded

Characteristics	Recorded data n (%)
Sex	
Women	31,301 (100.0%)
Screening round overall	
First round	3,524/31,301 (11.3%)
Follow-up	27,777/31,301 (88.7%)
Exam modality	
DM	17,333/31,301 (55.38%)
DBT	13,968/31,301 (44.62%)
DM screening round	
First round	1,862/17,333 (10.7%)
Follow-up	15,471/17,333 (89.3%)
DBT screening round	
First	1,662/13,968 (11.9%)
Follow-up	12,306/13,968 (88.1%)
Age at screening, years, median (interquartile range)	59 (54–64)
Age by groups	
49–54 years	9,647/31,301 (30.8%)
55–60 years	8,643/31,301 (27.6%)
61–66 years	8,105/31,301 (25.9%)
67–72 years	4,906/31,301 (15.7%)
Breast density (BI-RADS 5th edition)^a	
A (almost entirely fatty)	6,445/31,301 (20.6%)
B (scattered fibroglandular tissue)	14,569/31,301 (46.5%)
C (heterogeneously dense)	8,625/31,301 (27.6%)
D (extremely dense)	1,662/31,301 (5.3%)
Women diagnosed with breast cancer	252/31,301 (0.8%)

Numbers represent absolute values and percentages within parentheses, unless otherwise specified. ^aBreast Imaging Reporting and Data System (BI-RADS) classification.

The false positive rate (FPR) was higher in the AI strategy (4.8% (95% CI 4.5%, 5.0%)) than in the standard strategy (4.2% (95% CI 4.0%, 4.4%)), with an absolute difference of 0.6% (95% CI 0.3%, 0.9%).

On analyzing the results according to imaging modality (DM and DBT), in addition to a significant workload reduction (–62.1% and –65.5%, respectively), the results in terms of CDR and RR were different in each modality. In the AI strategy with DM, the CDR was 33.7% (95% CI 19.8%, 50.5%), higher than in the standard strategy (absolute difference of 1.6 of 1,000; 95% CI 0.9%, 2.4%). In addition, the RR was 28.2% (95% CI 20.2%, 36.3%), higher than in the standard strategy (absolute difference of 1.3%; 95% CI 1.0%, 1.8%). In the AI strategy with DBT, the CDR and RR were similar to the standard strategy. A relative difference of 0.9% (95% CI –10%, 11.8%), which is an absolute difference of 0.1 of 1,000 in the CDR, and a relative difference of –2.4% (95% CI –10.6%, 5.7%), which is an absolute difference of –0.1% in the RR, were observed. See Table 3 and Extended Data Figs. 1 and 2 for more details.

Characteristics of the cancers detected

In total, 252 cancers were detected (189 invasive and 63 carcinomas in situ): 198 via the standard strategy (158 invasive (79.8%) and 40 carcinomas in situ (20.2%)) and 228 cancers via the AI strategy (174 invasive (76.3%) and 54 carcinomas in situ (23.6%)). Thus, the AI strategy detected 10.1% more invasive carcinomas (95% CI 1.8%, 18.7%); 35% more carcinomas in situ (95% CI 7.6%, 60.3%); and a higher proportion of grade I invasive carcinomas (30.2% (95% CI 13.0%, 48.3%)), T1 (13.5%

(95% CI 3.2%, 24.1%)) and N0 invasive carcinomas (15.6% (95% CI 6.1%, 25.4%)) than the standard strategy, without observing significant differences in the molecular profile.

With DM, a total of 122 cancers were diagnosed (89 invasive and 33 carcinomas in situ). The AI strategy detected more invasive carcinomas, carcinomas in situ, grade I invasive carcinomas, and T1 and N0 invasive carcinomas than the standard strategy. The AI-based strategy showed an increase in the detection of luminal A tumors (44.4% (95% CI –6.5%, 88.3%)).

With DBT, a total of 130 cancers were diagnosed (100 invasive and 30 carcinomas in situ). No differences were found between the histopathological characteristics of cancers detected in both strategies in this modality.

Table 4 presents the overall characteristics of cancers detected by each strategy and modality.

Cancers missed for each strategy

Twenty-four cancers were detected only by the standard strategy, of which 23 of 24 (96%) were only recalled by one of the two radiologists. Of these, 11 cancers were identified by AI as low risk (scores 1–7) and were automatically labeled as normal in the AI strategy (9 of these low-risk cancers were DBT exams and 2 were DM exams). The other 13 cancers received scores of 8–10, but were not identified as cancer nor recalled by the human readers involved as part of the AI strategy. On the other hand, there were 54 screen-detected cancers only detected by the AI strategy, most of them (34 of 54 (63%)) were recalled by both radiologists.

Extended Data Table 1 shows the scores assigned by the AI system to the cancers detected. For more details, see Extended Data Tables 2–5.

Safety

No adverse events were reported in this study.

Discussion

The results from this prospective, paired, noninferiority clinical trial demonstrate that in a prospective setting it is safe to use AI to identify screening exams that can be automatically labeled as normal and avoid radiologist reading in a population screening program that uses both DM and DBT exams.

Overall, the AI strategy, which entailed radiologists reading only 36% of total exams, led to a 15.2% increase in the CDR from 6.3 of 1,000 to 7.3 of 1,000 and a 14.8% increase in the RR from 4.8% to 5.5%, while maintaining the PPV. Therefore, the CDR in the AI strategy was considered noninferior and statistically superior to that of the standard strategy and RR was not considered noninferior. Notably, cancer detection increased for both carcinomas in situ and invasive carcinomas, as the AI strategy detected 14 more carcinomas in situ and 16 more invasive carcinomas than the standard strategy. These results confirm that the high AI accuracy in identifying screening exams that are very likely normal together with a high cancer detection accuracy that provides effective concurrent decision support could be used to safely implement a partially autonomous AI screening workflow.

Interestingly, these prospective results are in line with early retrospective simulations investigating the use of such an AI strategy in which exams that were highly likely to be normal could be automatically assessed as normal^{4,25–27}. All found a potential safe reduction (no decrease in sensitivity or specificity) in the screening workload of 70%, 63%, 72% and 63%, respectively, in four distinct screening populations across Europe. Of note, our results include DBT screening, which is known to take at least twice as long to read as DM screening exams³. Therefore, the suggested safe workload reduction of 64% would be most impactful on radiologists' time in DBT screening programs.

Other prospective studies^{18–22} have investigated the introduction of AI in DM-based breast cancer screening. PRAIM²¹ was a multicenter study that compared the performance of AI-supported double reading (voluntarily chosen by radiologists) to standard double reading.

Table 2 | Screening performance outcomes in total, for each strategy

Screening performance indicator	AI strategy		Standard strategy		Absolute difference	Relative difference in percentages	Noninferiority/ superiority
	Proportion	Value ^a	Proportion	Value ^a	Value	Value %	P value
		(95% CI)		(95% CI)	(95% CI)	(95% CI)	
Main outcomes							
Screening readings (workload)	11,384/31,301	36.4	31,301/31,301	100	−19,917	−63.6	
		(35.8, 36.9)		(100,100)		(−64.2, −63.1)	
CDR	228/31,301	7.3	198/31,301	6.3	1	15.2	P<0.001 ^{bc}
		(6.3, 8.2)		(5.5, 7.2)	(0.4, 1.5)	(6.6, 24.4)	
RR	1,723/31,301	5.5	1,501/31,301	4.8	0.7	14.8	P=0.997 ^d
		(5.3, 5.8)		(4.6, 5.0)	(0.4, 1.0)	(9.0, 20.6)	
Secondary outcomes							
PPV of recall	228/1,723	13.23	198/1,501	13.19	0.04	0.31	P=0.500
		(11.63, 14.83)		(11.50, 14.90)	(−2.3, 2.4)	(−17.58, 18.03)	
FPR	1,495/31,049	4.8	1,303/31,049	4.2	0.6	14.7	P<0.001
		(4.5, 5.0)		(4.0, 4.4)	(0.3, 0.9)	(8.2, 21.3)	

For PPV, a z-test was used for testing the two-sided hypothesis of equality of proportions, while for FPR a paired McNemar test was used. For the CDR and RR, initially, one-tailed McNemar paired tests were applied for the noninferiority analyses. However, when noninferiority was achieved, the *P* value represented in the table is the one corresponding to the superiority analyses. ^aReadings are percentage of the total. CDR per 1,000 women screened. RR, PPV of recall and FPR per 100 women screened. ^bNoninferior. ^cSignificant difference. ^dNoninferiority is not demonstrated.

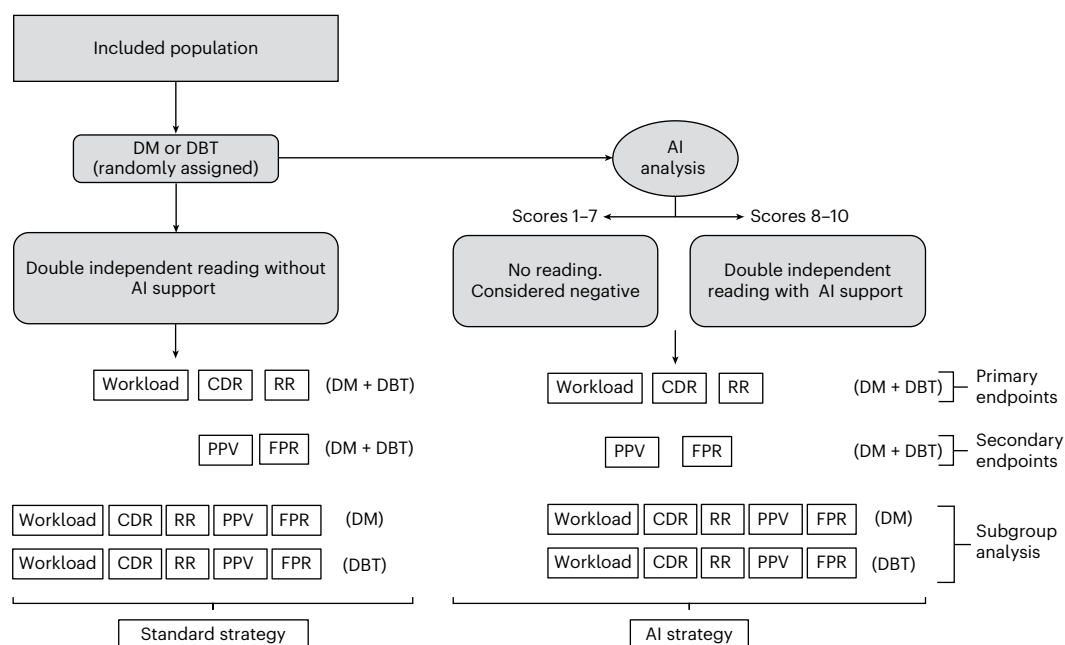


Fig. 2 | Flowchart describing the paired study design of the AITIC trial indicating the primary and secondary endpoints. Each participant's exam was read using two strategies: standard strategy (double human reading without AI support) and the AI-supported strategy, with double human reading with AI

support only for cases classified by the AI system with a score of 8 to 10. In this strategy, cases classified by the AI system with a score of 1 to 7 were considered low risk (approximately 70% of the studies) and would be automatically classified as normal.

It found an increase in the CDR of 17.6% in the AI group. It also analyzed a fictitious scenario in which studies triaged as normal by AI were not read by the radiologists. It showed that, in addition to a 56.7% reduction in the workload, the CDR remained 16.7% higher and the RR was statistically superior and 15% lower in the AI group than in the control group. On the other hand, MASAI^{18,19} and ScreenTrustCAD²⁰ introduced AI within the scope of reducing the workload. These studies were focused on totally or partially replacing the double reading of screening exams by single reading. Interestingly, MASAI, which maintained double reading of the

most suspicious exams with AI support (similar to our approach for the high-AI-risk exams) also reported an increase in the CDR (+20%) and a workload reduction of 44% without a negative impact on false positives. In a different design, ScreenTrustCAD investigated a direct replacement of the second radiologist reading by AI for all cases, without any AI support for the single reading, and found an impact on the CDR of +4%, a 50% workload reduction and a 21% increase in false positives prior to arbitration.

The benefits of double reading of high-AI-risk exams with AI support and the replacement of double reading by single reading in low-risk

Table 3 | Screening performance outcomes for each strategy in the subgroup analysis by modality (DM and DBT separately)

Screening performance indicator	AI strategy		Standard strategy		Absolute difference	Relative difference in percentages
	Proportion	Value ^a (95% CI)	Proportion	Value ^a (95% CI)	Value (95% CI)	Value (95% CI)
DM						
Screening readings (workload)	6,564/17,333	37.9 (37.2, 38.6)	17,333/17,333	100 (100, 100)	−10,769	−62.1 (−62.8, −61.4)
CDR	115/17,333	6.6 (5.4, 7.8)	86/17,333	5.0 (3.9, 6.0)	1.60 (0.9, 2.4)	33.7 (19.8, 50.5)
RR	1,082/17,333	6.2 (5.9, 6.6)	844/17,333	4.9 (4.5, 5.1)	1.30 (1.0, 1.8)	28.2 (20.2, 36.3)
PPV of recall	115/1,082	10.6 (8.8, 12.5)	86/844	10.2 (8.2, 12.2)	0.40 (−2.3, 3.2)	4.3 (−23.2, 31.1)
FPR	967/17,211	5.6 (5.3, 6.0)	758/17,211	4.4 (4.1, 4.7)	1.2 (0.8, 1.6)	27.6 (18.8, 36.5)
DBT						
Screening readings (workload)	4,820/13,986	34.5 (33.7, 35.3)	13,968/13,968	100 (100, 100)	−9,148	−65.5 (−66.3, −64.7)
CDR	113/13,968	8.1 (6.6, 9.6)	112/13,968	8.0 (6.5, 9.5)	0.1 (−2.0, 2.2)	0.9 (−10.0, 11.8)
RR	641/13,968	4.6 (4.2, 4.9)	657/13,968	4.7 (4.3, 5.0)	−0.1 (−0.5, 0.3)	−2.4 (−10.6, 5.7)
PPV of recall	113/641	17.6 (14.7, 20.6)	112/657	17.0 (14.2, 19.9)	0.6 (−3.5, 4.7)	3.1 (−20.8, 27.7)
FPR	528/13,838	3.82 (3.5, 4.2)	545/13,838	3.9 (3.6, 4.3)	−0.08 (−0.49, 0.25)	−3.4 (−12.7, 6.4)

The analyses presented in this table correspond to secondary outcomes. As the study was not powered for these objectives, CIs should be interpreted with caution. ^aReadings are the percentage of the total. CDR per 1,000 women screened. RR, PPV of recall and FPR per 100 women screened.

exams are also supported by the results of the observational study after AI implementation in the Capital Region of Denmark (increase in the CDR of 17%, reduction in false positives of 32% and reduction in workload of 33%)²⁸ using the same AI system as MASAI and AITIC.

Our results show a 14.8% increase in RR in the AI strategy. The study conducted by Ng et al.²² showed 0.16–0.30% additional recalls. Two other prospective studies (MASAI and ScreenTrustCAD)^{18–20} also showed an increase in the number of abnormal studies in the AI strategy, but both studies, unlike ours, held consensus meetings to assess these abnormal studies, which resulted in fewer women being recalled. The design of our screening program, which is without consensus, is associated with more recalls and may be more difficult to compare with studies conducted in screening programs with consensus. In addition, there are other factors in our study design that may also increase recalls, such as the fact that, in the AI reading strategy, AI-supported human double reading of cases with scores of 8–10 is performed, in contrast to other prospective studies where AI-supported double reading was only performed in cases with a score of 10^{18,19} or the AI information was not available during human reading or was on demand^{20,21}. Nevertheless, it is not possible to know the false negatives of nonrecalled women following the consensus of these studies.

Although the main objective of our study was not to compare results between DM and DBT, when looking at the impact of AI by modality (DM and DBT), in addition to the significant workload reduction (−62.1% and −65.5%, respectively), the results in terms of CDR and RR were different in each modality.

In the AI strategy with DM, the CDR was 33.7% higher than in the standard strategy and the RR was 28.2% higher than in the standard

strategy. In the AI strategy with DBT, the CDR and RR were similar to the standard strategy. A relative difference of 0.9% in the CDR and a relative difference of −2.4% in the RR were observed. The AI strategy with DM shows a substantial increase in the detection rate of both invasive carcinomas and carcinomas in situ as well as in the percentage of G1, T1 and N0 invasive carcinomas compared to the standard strategy. In addition, an increase in the detection of luminal A tumors was observed. MASAI¹⁹ shows similar results in terms of increased detection of invasive carcinomas in the AI strategy (24.4%), especially due to small tumors and those with lymph-node negative results, which should entail a positive impact on patient prognosis and lower aggressiveness and treatment costs. The increase in detection of carcinomas in situ in the AI with DM strategy is slightly higher than that observed in MASAI (66.7% and 51.1%, respectively). However, these authors showed greater carcinoma in situ detection (57.9%) in the AI strategy in their previous results¹⁸ and hypothesize that the radiologists' experience in working with AI and the feedback from the care of recalled women has probably influenced the later data. As in the MASAI series¹⁹, a significant percentage of the carcinomas in situ detected by the AI strategy (9 of 12) were intermediate or high grade. The RR was not shown to be noninferior. However, taking into account the increase in detection of 1.6 of 1,000, it can be claimed that this increase in recalls of 1.3% is acceptable.

In tomosynthesis, no impact on the detection or RR was demonstrated. Noninferiority was not able to be demonstrated, probably due to the sample size and threshold of the objective (noninferiority of −5%). The high screening performance with DBT in this program even without AI should be noted, likely due to the extensive experience of

Table 4 | Histological characteristics of detected cancers

Histological characteristics	Both strategies	AI strategy	Standard strategy	Absolute difference	Relative difference in percentages (95% CI)
All studies					
Total	252	228	198	30	15.2 (6.6, 23.8)
Invasiveness					
Invasive carcinoma	189	174	158	16	10.1 (1.8, 18.7)
In situ carcinoma	63	54	40	14	35.0 (7.6, 60.3)
Grade (invasive)					
Grade I	58	56	43	13	30.2 (13.0, 48.3)
Grade II	89	79	79	0	0 (-11.6, 11.6)
Grade III	42	39	36	3	8.3 (-9.1, 26.2)
Grade (in situ)					
Grade I	14	10	6	4	66.7 (-45.0, 151.6)
Grade II	26	23	18	5	27.8 (-9.0, 61.3)
Grade III	23	21	16	5	31.25 (-6.2, 65.2)
Tumor size^a					
T1	136	126	111	15	13.5 (3.2, 24.1)
T2	46	43	40	3	7.5 (-8.2, 23.8)
T3	6	5	6	-1	-20 (-67.6, 34.7)
T4	1	0	1	-1	-
Lymph node involvement (invasive)^a					
N0	150	141	122	19	15.6 (6.1, 25.4)
N+ (N1, N2, N3)	39	33	36	-3	-8.3 (-26, 9.1)
Molecular subtype (invasive)^b					
Luminal A	123	113	99	14	14.1 (-14.5, 41.7)
Luminal B Her2 ⁻	30	27	27	0	0.0 (-48.9, 48.9)
Luminal B Her2 ⁺	13	12	13	-1	-7.7 (-100, 56.8)
Her2 ⁺ (nonluminal)	6	5	6	-1	-20 (-67.6, 34.7)
Triple ⁻	6	6	5	1	20 (-80, 120)
Not assessable ^c	11	11	8	3	37.5 (-59.1, 137.5)
DM					
Total	122	115	86	29	33.7 (19.9, 47.6)
Invasiveness					
Invasive carcinoma	89	85	68	17	25 (11.5, 39.1)
In situ carcinoma	33	30	18	12	66.7 (22.3, 103.3)

Table 4 (continued) | Histological characteristics of detected cancers

Histological characteristics	Both strategies	AI strategy	Standard strategy	Absolute difference	Relative difference in percentages (95% CI)
Grade (invasive)					
Grade I	29	28	18	10	55.6 (21.1, 86.7)
Grade II	35	32	30	2	6.7 (-13.4, 26.9)
Grade III	25	25	20	5	25 (5.0, 48.9)
Grade (in situ)					
Grade I	7	5	2	3	150 (-98.8, 292.5)
Grade II	13	12	6	6	100 (8.6, 158.8)
Grade III	13	13	10	3	30 (-6.5, 65.3)
Tumor size^a					
T1	65	63	50	13	26 (11.2, 42.0)
T2	19	18	13	5	38.43 (-1.7, 74.3)
T3	4	4	4	0	0 (-49.0, 49.0)
T4	1	0	1	-1	-
Lymph node involvement (invasive)^a					
N0	75	73	55	18	32.7 (18.0, 48.6]
N+ (N1, N2, N3)	14	12	13	-1	-
Molecular subtype (invasive)^b					
Luminal A	54	52	36	16	44.4 (-6.5, 88.3)
Luminal B Her2 ⁻	17	15	16	-1	-6.3 (-63.8, 54.1)
Luminal B Her2 ⁺	7	7	7	0	0 (-100, 100)
Her2 ⁺ (nonluminal)	3	3	3	0	0 (-100, 100)
Triple ⁻	4	4	3	1	33.3 (-93.7, 133.3)
Not assessable ^c	4	4	3	1	33.3 (-93.7, 133.3)
DBT					
Total	130	113	112	1	8.9 (-9.7, 11.5)
Invasiveness					
Invasive carcinoma	100	89	90	-1	-1.1 (-11.6, 9.3)
In situ carcinoma	30	24	22	2	9.1 (-24.7, 41.8)
Grade (invasive)					
Grade I	29	28	25	3	12 (-7.3, 32.6)
Grade II	54	47	49	-2	-4.1 (-19.0, 10.6)
Grade III	17	14	16	-2	-12.5 (-39.5, 15.1)
Grade (in situ)					
Grade I	7	5	4	1	25 (-76.1, 110.5)
Grade II	13	11	12	-1	-8.3 (-40.6, 24.9)
Grade III	10	8	6	2	33.3 (-46.4, 98.5)

Table 4 (continued) | Histological characteristics of detected cancers

Histological characteristics	Both strategies	AI strategy	Standard strategy	Absolute difference	Relative difference in percentages (95% CI)
Tumor size^a					
T1	71	63	61	2	3.3 (−10.9, 17.5)
T2	27	25	27	−2	−7.4 (−23.4, 6.0)
T3	2	1	2	−1	−50 (−90.6, 48.6)
T4	0	–	–	–	–
Lymph node involvement (invasive)^a					
N0	75	68	67	1	1.5 (−10.5, 13.6)
N+ (N1, N2, N3)	25	21	23	−2	−8.7 (−31.6, 14.2)
Molecular subtype (invasive)^b					
Luminal A	69	61	63	−2	−3.2 (−36.3, 30.3)
Luminal B Her2 ⁺	13	12	11	1	9.1 (−64.5, 77.8)
Luminal B Her2 [−]	6	5	6	−1	−16.7 (−100, 67.2)
Her2 ⁺ (nonluminal)	3	2	3	−1	−33.3 (−100, 71.1)
Triple [−]	2	2	2	0	0 (−100, 100)
Not assessable ^c	7	7	5	2	40 (−75.3, 140)

Her2⁺, human epidermal growth factor receptor 2. ^aTumour, node and metastasis staging of detected cancers was based on the Union for International Cancer Control's classification system, 8th edition²⁹. ^bMolecular subtype classification based on the World Health Organization's *WHO Classification of Tumours*, 5th edition³⁰. ^cNot able to be assessed due to lack of information from one or more receptors, usually patients undergoing surgery at another center. The analyses presented in this table correspond to secondary outcomes. As the study was not powered for these objectives, CIs should be interpreted with caution.

the radiologists reading DBT (median of 5 years of experience reading DBT screening tests). AI systems may perform differently for DBT than for DM and may be more accurate for DM when compared to the average radiologist performance⁹. However, the results in the detection and RRs achieved by the AI-DBT strategy (8.1 of 1,000 and 4.6%, respectively) are superior to those achieved by the AI-DM strategy (6.6 of 1,000 and 6.2%, respectively). Nevertheless, the significant reduction in workload achieved (65.5%) is the most important aspect in favor of incorporating AI in screening programs using DBT. Thus, the strengths of this work are that it is a prospective, paired study that aimed to provide information on the use of AI in a real-life screening setting with DBT, allowing for assessing nonhuman reading of a significant percentage of studies classified as low risk by AI.

One of the major barriers for clinical adoption of our proposed strategy is certainly the ethical challenges arising from having most screening mammograms read only by AI and automatically assessed as normal without radiologists' involvement. This could lead to cancers being missed (11 cancers in our study), but, as demonstrated in our study, there are more cancers being missed by radiologists when not using AI for decision support in screening (54 cancers in our study). Furthermore, among the cancers with a score <8 missed by the AI strategy, most (91%) were only detected by one of the two readers in the standard strategy, indicating their potential subtle appearance. Nevertheless, the ethical implications of relying on AI for initial reads versus radiologists must be carefully considered to ensure patient safety and trust in screening processes. If there is no radiologist review of a significant proportion of exams, additional screening quality assurance processes such as automated mammography image quality

control and continuous postmarket surveillance of AI performance are necessary steps before implementing such an autonomous AI screening workflow for the very likely normal screening exams.

Our study has the limitation of being a single-site investigation with a screening workflow of double reading without arbitration by expert radiologists in breast imaging and breast screening who have several years of experience in the use of AI. Further studies are needed to validate these findings in other settings. In addition, mammography images were acquired using four devices, all from a single vendor. Results were also obtained with one breast AI system; extrapolation to other AI systems must be evaluated further, given the significant differences in AI accuracy, robustness and user interface across commercial systems. The sample size was calculated to include all studies for DM and DBT together, reflecting our usual practice, and not for DM and DBT separately. Consequently, the results in DBT have been less consistent. Finally, as it is a paired design, it is not possible to know the interval carcinoma rate for each strategy.

Further research is needed on the legal and safety considerations involved in preventing the reading of low-risk studies, as well as on the performance of AI in DBT.

In conclusion, AI triage and AI-supported screening that excludes low-risk mammograms from radiologist reading could be a safe and effective screening strategy that would allow for a substantial reduction in reading workload in breast cancer screening programs without negatively affecting cancer detection or the PPV of the recalled studies.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information,

acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-026-04277-x>.

References

- Caswell-Jin, J. L. et al. Analysis of breast cancer mortality in the US—1975 to 2019. *JAMA* **331**, 233–241 (2024).
- Lee, C. I. et al. National performance benchmarks for screening digital breast tomosynthesis: update from the Breast Cancer Surveillance Consortium. *Radiology* **307**, e222499 (2023).
- Romero Martín, S. et al. Prospective study aiming to compare 2D mammography and tomosynthesis + synthesized mammography in terms of cancer detection and recall. From double reading of 2D mammography to single reading of tomosynthesis. *Eur. Radiol.* **28**, 2484–2491 (2018).
- Raya-Povedano, J. L. et al. AI-based strategies to reduce workload in breast cancer screening with mammography and tomosynthesis: a retrospective evaluation. *Radiology* **300**, 57–65 (2021).
- Elías-Cabot, E., Romero-Martín, S., Raya-Povedano, J. L., Brehl, A. K. & Álvarez-Benito, M. Impact of real-life use of artificial intelligence as support for human reading in a population-based breast cancer screening program with mammography and tomosynthesis. *Eur. Radiol.* **34**, 3958–3966 (2024).
- van Winkel, S. L. et al. Impact of artificial intelligence support on accuracy and reading time in breast tomosynthesis image interpretation: a multi-reader multi-case study. *Eur. Radiol.* **31**, 8682–8691 (2021).
- Rodríguez-Ruiz, A. et al. Detection of breast cancer with mammography: effect of an artificial intelligence support system. *Radiology* **290**, 305–314 (2019).
- Pacilè, S. et al. Improving breast cancer detection accuracy of mammography with the concurrent use of an artificial intelligence tool. *Radiol. Artif. Intell.* **2**, e190208 (2020).
- Romero-Martín, S. et al. Stand-alone use of artificial intelligence for digital mammography and digital breast tomosynthesis screening: a retrospective evaluation. *Radiology* **302**, 535–542 (2022).
- Rodríguez-Ruiz, A. et al. Stand-alone artificial intelligence for breast cancer detection in mammography: comparison with 101 radiologists. *J. Natl Cancer Inst.* **111**, 916–922 (2019).
- Salim, M. et al. External evaluation of 3 commercial artificial intelligence algorithms for independent assessment of screening mammograms. *JAMA Oncol.* **6**, 1581–1588 (2020).
- McKinney, S. M. et al. International evaluation of an AI system for breast cancer screening. *Nature* **577**, 89–94 (2020).
- Larsen, M., Aglen, C. F., Hoff, S. R., Lund-Hanssen, H. & Hofvind, S. Possible strategies for use of artificial intelligence in screen-reading of mammograms, based on retrospective data from 122,969 screening examinations. *Eur. Radiol.* **32**, 8238–8246 (2022).
- Shoshan, Y. et al. Artificial intelligence for reducing workload in breast cancer screening with digital breast tomosynthesis. *Radiology* **303**, 69–77 (2022).
- Lång, K. et al. Identifying normal mammograms in a large screening population using artificial intelligence. *Eur. Radiol.* **31**, 1687–1692 (2021).
- Rodríguez-Ruiz, A. et al. Can we reduce the workload of mammographic screening by automatic identification of normal exams with artificial intelligence? A feasibility study. *Eur. Radiol.* **29**, 4825–4832 (2019).
- Dahlblom, V., Dustler, M., Tingberg, A. & Zackrisson, S. Breast cancer screening with digital breast tomosynthesis: comparison of different reading strategies implementing artificial intelligence. *Eur. Radiol.* **33**, 3754–3765 (2023).
- Lång, K. et al. Artificial intelligence—supported screen reading versus standard double reading in the Mammography Screening with Artificial Intelligence trial (MASAI): a clinical safety analysis of a randomised, controlled, non-inferiority, single-blinded, screening accuracy study. *Lancet Oncol.* **24**, 936–944 (2023).
- Hernström, V. et al. Screening performance and characteristics of breast cancer detected in the Mammography Screening with Artificial Intelligence trial (MASAI): a randomised, controlled, parallel-group, non-inferiority, single-blinded, screening accuracy study. *Lancet Digit. Health* **7**, e175–e183 (2025).
- Dembrower, K. et al. Artificial intelligence for breast cancer detection in screening mammography in Sweden: a prospective, population-based, paired-reader, non-inferiority study. *Lancet Digit. Health* **5**, e703–e711 (2023).
- Eisemann, N. et al. Nationwide real-world implementation of AI for cancer detection in population-based mammography screening. *Nat. Med.* **31**, 917–924 (2025).
- Ng, A. Y. et al. Prospective implementation of AI-assisted screen reading to improve early detection of breast cancer. *Nat. Med.* **29**, 3044–3049 (2023).
- European Guidelines for Breast Cancer Screening. *European Commission* (May 2023); <https://cancer-screening-and-care.jrc.ec.europa.eu/en/ecibc/european-breast-cancer-guidelines?topic=65&usertype>
- Elías-Cabot, E., Romero-Martín, S., Raya-Povedano, J. L., Rodríguez-Ruiz, A. & Álvarez-Benito, M. Artificial Intelligence in Breast Cancer Screening Program in Cordoba (AITIC). *Zenodo* <https://doi.org/10.5281/zenodo.17625633> (2025).
- Lauritzen, A. D. et al. An artificial intelligence-based mammography screening protocol for breast cancer: outcome and radiologist workload. *Radiology* **304**, 41–49 (2022).
- Dembrower, K. et al. Effect of artificial intelligence-based triaging of breast cancer screening mammograms on cancer detection and radiologist workload: a retrospective simulation study. *Lancet Digit. Health* **2**, e468–e474 (2020).
- Leibig, C. et al. Combining the strengths of radiologists and AI for breast cancer screening: a retrospective analysis. *Lancet Digit. Health* **4**, e507–e519 (2022).
- Lauritzen, A. D. et al. Early indicators of the impact of using AI in mammography screening for breast cancer. *Radiology* **311**, e232479 (2024).
- Brierley, J., Gospodarowicz, M. D. & Wittekind, C. T. *TNM Classification of Malignant Tumours International Union Against Cancer* (Wiley, 2017).
- WHO Classification of Tumours. *Breast Tumours* Vol. 2 (WHO, 2019).

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Methods

Ethics approval

All patients provided written informed consent before enrollment. This prospective clinical trial was compliant with the Health Insurance Portability and Accountability Act and the design was preregistered at <https://classic.clinicaltrials.gov/ct2/show/NCT04949776>. In March 2021, the study protocol and the informed consent received a favorable ruling from the Institutional Review Board (IRB) at Reina Sofía University Hospital of Córdoba Research Ethics Committee (IRB No 4932).

Clinical trial design and patients

This is a prospective paired noninferiority experimental study.

The individual deidentified participant dataset, data dictionary defining each field and study protocol are publicly accessible via Zenodo at <https://doi.org/10.5281/zenodo.17625633> (ref. 24).

This clinical trial was carried out in the Córdoba Breast Cancer Screening Unit, Córdoba, Spain, as part of the Andalusian screening program in Spain. Women participants aged 50 to 71 years old (including women who reach that age in the year of appointment) are invited to participate in the screening program at 2-year intervals (ClinicalTrials.gov ID: [NCT04949776](https://clinicaltrials.gov/ct2/show/study/NCT04949776)).

All women participating in the screening program from 15 March 2022 to 11 January 2024 were eligible and invited to participate in this clinical trial. They were asked to sign an informed consent form. For screening purposes, sex and age data were defined by reference to the population census. All women aged 50 to 71 years, regardless of socioeconomic status, were considered.

After signing informed consent, women were excluded from the clinical trial if they had symptoms or signs of suspected breast cancer; if they had breast prostheses; or if their images could not be processed by the AI system (for example, because of the presence of visible breast implants or because of erroneous PACS image transfer).

To investigate the hypothesis and to account for ethical aspects, this clinical trial had a paired design (Fig. 2). Each participant received both reading strategies:

- (1) The standard of care: double human reading without AI support.
- (2) The AI strategy: double human reading with AI support (by two additional radiologists) only for cases classified by the AI system with a score of 8 to 10 (approximately 30% of the studies most likely to have cancer). Cases classified by the AI system with a score of 1 to 7 were considered low risk (approximately 70% of the studies) and automatically classified as normal.

Procedures

Each participant underwent a standard four-view DM or DBT exam based on resource availability, according to the standard of care. There were no rules governing imaging modality selection. The assignment of women to the different types of mammography equipment was not changed for this study. Images were acquired using four devices: three DM devices (Lorad Selenia, Hologic) and one DBT device (3Dimensions, Hologic).

All exams were independently double read without consensus or arbitration. Screening readings were randomly assigned according to availability of the nine dedicated breast radiologists in the hospital's radiology department (3 to 21 years of experience in breast cancer screening), regardless of whether they were readings with or without AI assistance and regardless of the reader's experience. The mammogram offered to the reader was the oldest one performed that had not been read by that same reader. In each reading, the radiologist noted the breast density (measured subjectively according to the BI-RADS classification³¹), the presence of findings and their location in the breast, the assigned BI-RADS category (1 to 5) and the decision whether or not to recall. Women were recalled if any reader decided to recall.

Women received a single result (recall or no recall) once all readings were completed. If recalled, additional exams (special mammography views, tomosynthesis, contrast-enhanced mammography and/or ultrasonography) and, when required, biopsy were performed at the women's referral unit. The screening program did not include ultrasonography or other additional examinations for women with high breast density. No adverse events were reported in this study.

This clinical trial used a commercially available AI system for breast cancer detection, Transpara (version 1.7 ScreenPoint Medical), which had been used for previous research studies by the same group of radiologists involved in this study.

The AI system's performance, generalizability and clinical utility has been previously investigated in over 30 other peer-reviewed scientific publications. Most significantly, researchers have shown that this system can achieve stand-alone breast cancer detection performances comparable to radiologists¹⁰, that radiologists become more accurate when using such an AI system for concurrent decision support reading mammograms, leading to higher screening CDRs^{18,19}, and that it is safe and effective to implement in breast cancer screening to replace the need for the double reading^{18,19}.

The AI system automatically detects and classifies regions suspicious of breast cancer in radiological images that adhere to the input compatibility criteria of the system and with standard Digital Imaging and Communications in Medicine. This version of the system can be used in combination with DM and DBT images acquired with systems from all major mammography equipment manufacturers (Siemens Healthineers, Hologic, General Electric, Giotto, Planmed, Fujifilm). The system can analyze any number of standard (craniocaudal, mediolateral oblique) and nonstandard (exaggerated, lateral) views per breast. Images of women with breast implants are not compatible unless the implant has been displaced during compression.

The AI system outputs per exam are (1) explainable image overlay markings for regions suspicious of breast cancer per view, alongside a suspiciousness score per region between 1 and 100 and an indication of the finding morphological type, and (2) an exam-based indication of the risk that the exam contains a visible lesion suspicious of breast cancer that requires additional follow-up (three categories: 1–7 low risk, 8–9 intermediate risk, 10 elevated risk).

The system uses deep convolutional neural networks as backbone algorithms to analyze DM and DBT images and detect lesions suspicious for breast cancer. Convolutional neural networks are state-of-the-art machine learning tools for image classification. Different convolutional neural networks are trained with different architectures and training datasets for malignancy detection of different morphological subtypes and taking a variable number of images as input.

The AI system was developed using a large-scale heterogeneous curated database of more than 15 million two-dimensional and three-dimensional X-ray breast images acquired in real-world breast cancer screening programs and diagnostic clinics. The data originated from over 15 sites across 10 countries across North America, Europe and Asia, including over 15,000 malignant cases (pathology proven). Negative cases were confirmed through multiple years of follow-up.

The AI system software version and operating points used in the present evaluation were established before the clinical trial. The choice of a score of 7 as the cutoff point for indicating that human reading was not necessary was based on previous studies conducted by this research team and other published research^{4,5,25}. None of the clinical data used in this clinical trial was used in any aspect of algorithm development.

The human readers in the AI strategy had available the information provided by the AI system (marks according to the type of lesion with a score according to the probability of having cancer and global score of the study), but the radiologist was the person who ultimately made the decision to recall a woman or not.

Data on the procedures, outcomes, participants and detected cancers were retrieved from the medical records. Race and ethnicity

were not individually recorded, but the majority of the target population was Caucasian.

Safety. The study protocol was reviewed and approved by the institutional ethics committee, which determined that the study posed minimal risk to participants.

Safety monitoring focused on nonphysical risks potentially associated with the use of AI-based image analysis, including data integrity, patient privacy and the potential for diagnostic error. All mammographic images were fully anonymized prior to analysis, and data handling complied with applicable data protection regulations. Any unexpected incidents related to data processing, algorithm malfunction or breaches of confidentiality were predefined as reportable events and were monitored throughout the study period.

Outcomes. The primary outcomes were workload, CDR and RR calculated globally. Workload was computed as the absolute number of radiologist readings in each strategy. A cancer detected during the diagnostic work-up after a recall and confirmed by pathology was considered the gold standard.

The CDR was estimated as the number of cancers detected per 1,000 screening examinations for each strategy. Cases that were not recalled or that were recalled with malignancy not demonstrated on complementary imaging tests or biopsies were considered negative. Follow-up lasted for a minimum of 180 days after screening mammography. For each case, a strategy was positive if one or both readings included referral.

As secondary objectives, PPV of recalls and FPR were assessed globally. A subgroup analysis by modality was performed and workload, CDR, RR, PPV of recalls and FPR were calculated for DM and for DBT separately. The PPV of recalls was calculated as the number of cancers diagnosed per 100 women recalled for each strategy. The FPR was calculated as the number of women recalled who were not finally diagnosed with cancer for each strategy per 100 negative screening examinations.

Sample size

For the sample size calculation, the method by Connor³² for paired studies was used, taking into account the sample size necessary to demonstrate noninferiority (5% margin) in terms of sensitivity for cancer detection via McNemar's z-test (one-sided). An alpha (type I error estimate) value of 5% and a beta (type II error estimate) of 20% (representing a power for the study of 80%, $1 - \beta$) was used. In addition, we used the parameters of an assumed cancer incidence of 6 of 1,000 (value of cancer incidence when AI is not used in screening based on initial experience and more recently verified in ref. 5). We also calculated how many cancers would be detected by the standard strategy alone and how many would be detected by the AI strategy proposed in this study using the database and methods used in ref. 4 but using Transpara software version 1.7.0 (as opposed ref. 4, which used version 1.6.0, which has inferior algorithms). In the results obtained, 1.8% of cancers were detected using the standard strategy but not using the proposed strategy with AI, while the opposite occurred in 9% of cancers. With these parameters, the sample size needed was defined as 27,000 women in order to demonstrate the noninferiority of the AI strategy over the standard strategy globally, including both DM and tomosynthesis studies.

The three coprimary endpoints were CDR, RR and workload. It is worth mentioning that no formal multiplicity correction was applied between them for the sample size calculation. Readers are encouraged to interpret the findings in the context of these coprimary outcomes.

Statistical analyses

The hypothesis raised in this work is that a partially autonomous AI-supported screening strategy (that is, no human reading in cases

with low-AI risk (AI strategy)) would allow for reducing the workload of a screening program that uses both DM and DBT exams compared to the standard-of-care reading without demonstrating inferiority in terms of the CDR and RR of the program. Descriptive analyses of the variables considered in the study were carried out in the complete dataset of women. Categorical variables were expressed as absolute and relative frequencies, while continuous variables were expressed as median and interquartile range. To compare the AI-assisted strategy with the standard-of-care strategy, we evaluated the CDR and RR, as well as secondary metrics including the PPV of recall and FPR. For all variables, percentages were calculated for each strategy, along with corresponding 95% CI. Differences between AI and standard strategies were assessed using Wald statistics with or without the Bonett–Laplace adjustment³³, as appropriate.

For the primary outcomes (CDR and RR), we conducted noninferiority testing using the following approach: reference values for the CDR and RR were identified from previously published noninferiority studies and screening benchmarks in the literature. Based on clinical consensus, a 5% relative tolerance margin over those values was considered acceptable for declaring noninferiority. For the CDR (where higher values are better), a noninferiority margin was defined as a 5% relative reduction from the reference value. For the RR (where lower values are generally preferred), the noninferiority margin was set as a 5% relative increase from the reference value.

These thresholds were used as the minimum acceptable performance levels for the AI-assisted strategy. One-sided 97.5% CIs were established for the difference in proportions (AI standard) for the CDR and RR. These were derived from two-sided 95% CIs. Noninferiority was declared if the lower limit of the one-sided CI exceeded the predefined noninferiority margin. Visual representation of these thresholds is shown in Extended Data Figs. 1 and 2.

If noninferiority was demonstrated for a given outcome, we proceeded to test for superiority using a paired exact McNemar test. For PPV and FPR, noninferiority was not formally tested. Instead, standard two-sided hypothesis tests were conducted: a z-test was used for comparing PPV (assuming independent data) and a paired McNemar test was used for comparing FPR.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Individual deidentified participant dataset, data dictionary defining each field, protocol and informed consent form-patients information sheet are publicly accessible via Zenodo at <https://doi.org/10.5281/zenodo.17625633> (ref. 24). Access ends 10 years following article publication.

Code availability

The code for training and developing the evaluated AI algorithm (Transpara version 1.7, ScreenPoint Medical) is part of a proprietary system. The commercial AI system is available for external research evaluation collaborations for researchers who provide a relevant and methodologically sound proposal. Proposals should be directed to alejandro.rodriguezruiz@screenpointmed.com and will be answered within 1 month. We provide a technical description of the backbone algorithm of the AI system in the Methods. The software used to perform the statistical analyses in this study was R, version 4.4.2. The main open-source libraries used for the analyses were: ggplot2 (for graphic purposes) and PropCIs (for confidence intervals estimation).

References

31. D'Orsi C. J. et al. *ACR BI-RADS Atlas, Breast Imaging Reporting and Data System* (American College of Radiology, 2013).

32. Connor, R. J. Sample size for testing differences in proportions for the paired-sample design. *Biometrics* **43**, 207–211 (1987).
33. Bonett, D. G. & Price, R. M. Adjusted Wald confidence interval for a difference of binomial proportions based on paired data. *J. Educ. Behav. Stat.* **37**, 479–488 (2012).

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Author contributions

E.E.-C. and M.A.-B. conceptualized the study and wrote the study protocol with input from S.R.-M., J.L.R.-P. and A.R.-R. E.E.-C., S.R.-M. and J.L.R.-P. contributed to clinical data collection. E.E.-C. and M.A.-B. contributed to data analysis. S.R.-M., J.L.R.-P. and A.R.-R. contributed to data interpretation. E.E.-C. wrote the first draft of the report. All authors reviewed the paper and provided important intellectual

content. All authors discussed the results and contributed to the final paper. All authors read and approved the manuscript.

Competing interests

The study was not funded by Industry. A.R.R. is an employee of ScreenPoint Medical. The other authors declare no competing interests.

Additional information

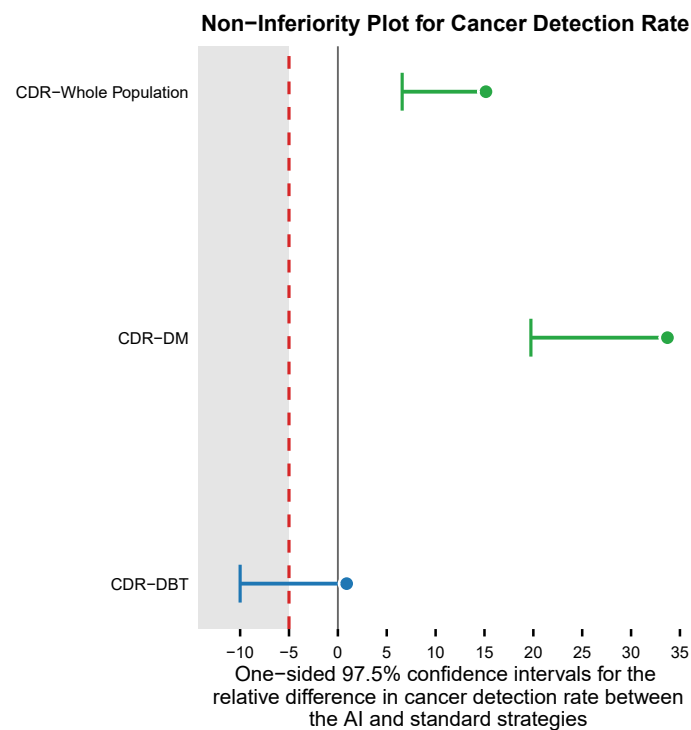
Extended data is available for this paper at <https://doi.org/10.1038/s41591-026-04277-x>.

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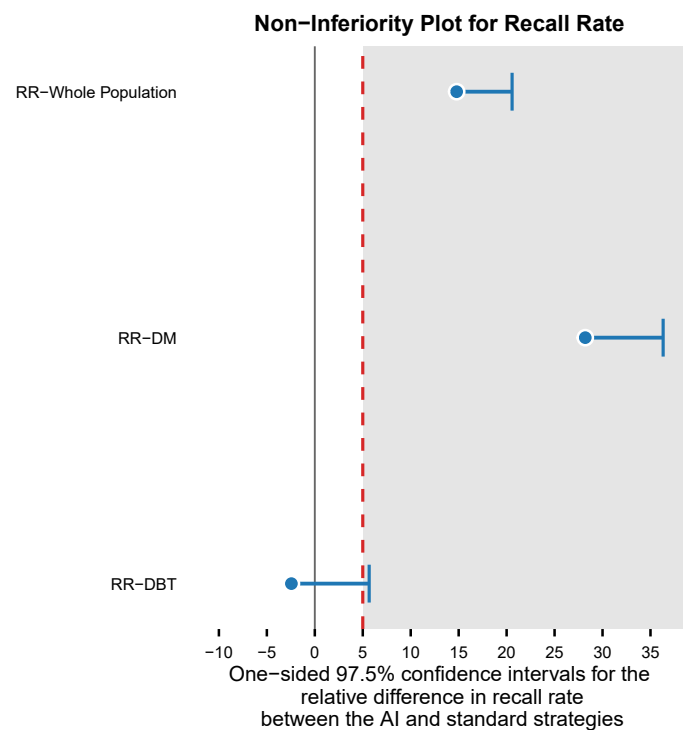
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Extended Data Fig. 1 | Non-inferiority plots for CDR. The horizontal axis is measured in percentage units of cancer detection. The dots of the horizontal lines represent the relative difference in CDR between both strategies. The whiskers represent the lower limits of the confidence intervals. The dashed vertical red line represents the non-inferiority threshold. Green indicates 'non-

inferiority achieved' and blue indicates 'non-inferiority not confirmed'. Relative differences in CDR between the AI strategy and standard strategy for the whole population (N = 31301): 15.2 (6.6, 24.4), for DM (N = 17333): 33.7 (19.8, 50.5), and for DBT (N = 13968): 0.9 (-10.0, 11.8). DM: digital mammography; DBT: digital breast tomosynthesis; CDR: cancer detection rate; AI: artificial intelligence.



Extended Data Fig. 2 | Non-inferiority plots for RR. The horizontal axis is measured in percentage units of recall. The dots of the horizontal lines represent the relative difference in RR between both strategies. The whiskers represent the lower limits of the confidence intervals. The dashed vertical red line represents the non-inferiority threshold. Green indicates 'non-inferiority achieved' and blue

indicates 'non-inferiority not confirmed'. Relative difference in RR between the AI strategy and standard strategy for the whole population ($N = 31301$): 14.8 (9.0, 20.6), for DM ($N = 17333$): 28.2 (20.2, 36.3), and for DBT ($N = 13968$): -2.4 (-10.6, 5.7). DM: digital mammography; DBT: digital breast tomosynthesis; RR: recall rate; AI: artificial intelligence.

Extended Data Table 1 | Score assigned by the artificial intelligence system to the detected cancers

Score	ALL STUDIES (n=252 cancers)	DM STUDIES (n=122 cancers)	DBT STUDIES (n=130 cancers)
1	1	1	0
2	0	0	0
3	1	0	1
4	1	0	1
5	1	0	1
6	4	0	4
7	3	1	2
8	12	4	8
9	13	6	7
10	216	110	106

DM, digital mammography. DBT, digital breast tomosynthesis.

Extended Data Table 2 | Cancers detected and not detected by both strategies (all studies)

	AI strategy +	AI strategy -	
Standard strategy +	174	24	198
Standard strategy -	54	0	54
	228	24	

AI: artificial intelligence

The table shows the number of cancers detected (+) and not detected (-) by the standard strategy, by the AI strategy, and by both strategies. All exams were included.

Extended Data Table 3 | Cancers detected and not detected by both strategies (Digital Mammography studies)

	AI strategy +	AI strategy -	
Standard strategy +	79	7	86
Standard strategy -	36	0	36
	115	7	

AI: artificial intelligence

The table shows the number of cancers detected (+) and not detected (-) with Digital Mammography by the standard strategy, by the AI strategy and by both strategies.

Extended Data Table 4 | Cancers detected and not detected by both strategies (Digital Breast Tomosynthesis studies)

	AI strategy +	AI strategy -	
Standard strategy +	95	17	112
Standard strategy -	18	0	18
	113	17	

AI: artificial intelligence

The table shows the number of cancers detected (+) and not detected (-) with Digital Breast Tomosynthesis by the standard strategy, by the AI strategy, and by both strategies.

Extended Data Table 5 | Characteristics of cancers with scores 1–7 (low risk), not detected by the AI strategy

Score	Histologic type	Morphologic type	Grade	Recalled by one or two radiologists	Modality
7	Invasive lobular carcinoma	Asymmetry	I	one	DM
7	Ductal carcinoma in situ	Architectural distortion	II	one	DBT
7	Invasive ductal carcinoma	Calcifications	II	one	DBT
6	Invasive ductal carcinoma	Mass	III	one	DBT
6	Invasive ductal carcinoma	Mass	II	one	DBT
6	Invasive ductal carcinoma	Asymmetry	II	one	DBT
6	Ductal carcinoma in situ	Calcifications	I	one	DBT
5	Invasive ductal carcinoma	Asymmetry	II	one	DBT
4	Invasive ductal carcinoma	Architectural distortion	II	one	DBT
3	Ductal carcinoma in situ	Calcifications	III	one	DBT
1	Ductal carcinoma in situ	Mass	I	two	DM

DM, digital mammography. DBT, digital breast tomosynthesis.

The table shows the histopathological type, radiological presentation, tumor grade, visualization by one or two radiologists, and modality for carcinomas with a score <8 not detected by the AI strategy.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | | |
|-------------------------------------|--|
| n/a | Confirmed |
| ☐ | ☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☐ | ☒ A description of all covariates tested |
| ☒ | ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Excel software (version 2511) was used to collect the data, available in Zenodo Repository https://doi.org/10.5281/zenodo.17625633 .
Data analysis	The code for training and developing the evaluated AI algorithm (Transpara® version 1.7, ScreenPoint Medical) is part of a proprietary system. The commercial AI system is available for external research evaluation collaborations for researchers who provide a relevant and methodologically sound proposal. Proposals should be directed to alejandro.rodriquezruiz@screenpointmed.com and will be answered within 1 month. We provide a technical description of the backbone algorithm of the AI system in the online Methods. All statistical analyses were performed with R software25, versión 4.3.2.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Individual de-identified participant dataset, data dictionary defining each field, and protocol are publicly accessible via Zenodo at <https://doi.org/10.5281/zenodo.17625633>. Access ends ten years following article publication.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

This clinical trial has been carried out in the Córdoba Breast Cancer Screening Program (CBCSP) which is part of the Andalusian screening program in Spain. Women aged 50-71 years old (including women who reach that age in the year of appointment) are invited to participate in this program. For screening purposes, sex and age data were defined by reference to the population census.

Reporting on race, ethnicity, or other socially relevant groupings

Race/ethnicity was not individually recorded, but the majority of the target population was of caucasian race. All women aged 50-71 years, regardless of their socio-economic status, were considered.

Population characteristics

Any woman aged between 50 and 71 years old (including women who reach that age in the year of appointment) eligible to participate in the Breast Cancer Screening Program (Cordoba, Spain) that have been randomly assigned in the Hologic equipment (DM or DBT) was invited to participate in this clinical trial and asked to sign informed consent. After signing informed consent, women were excluded from the clinical trial if they had symptoms or signs of suspected breast cancer; if they had breast prostheses, or if their images could not be processed by the AI system (e.g. because of the presence of visible breast implants, because of erroneous PACS image transfer). Characteristics of the population included in the study are listed in Table 1.

Recruitment

Women who consecutively participate in the Breast Cancer Screening Program in Córdoba from March 15th 2022 to January 11th 2024

Ethics oversight

The study protocol has been approved by the Córdoba provincial ethics committee and the participating women have signed and informed consent.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

For the sample size calculation, the method by Connor R. J. 1987 29 for paired studies was used, taking into account the sample size necessary to demonstrate non-inferiority (5% margin)/superiority in terms of sensitivity for cancer detection via McNemar's Z-test (one-sided). An alpha (type I error estimate) value of 5% and a beta (type II error estimate) of 20% (representing a power for the study of 80%, 1 - beta) was used. In addition, we used the parameters of an assumed cancer incidence of 6/1000 (value of cancer incidence when AI is not used in screening based on initial experience and more recently verified in Elias-Cabot et al 2023). We also calculated how many cancers would be detected by the standard strategy alone and how many would be detected by the AI strategy proposed in this study using the database and methods used in the Radiology journal article ("AI-based strategies to reduce workload in breast cancer screening with digital mammography and digital breast tomosynthesis: a retrospective evaluation of data collected from the Tomosynthesis Cordoba Screening Trial." RAD-20-3555) but using Transpara® software version 1.7.0 (as opposed to the original article, which used version 1.6.0, which has inferior algorithms). In the results obtained, 1.8% of cancers were detected using the standard strategy but not using the proposed strategy with AI, while the opposite occurred in 9% of cancers. With these parameters, the sample size needed was estimated to be at least 27,000 women in order to demonstrate the non-inferiority and/or superiority of artificial intelligence over standard strategy, globally, including both digital mammography and tomosynthesis studies.

	From March 15th 2022 to January 11th 2024, 31,856 consecutive women attending the breast cancer screening program agreed to participate in this trial.
Data exclusions	In total, 555 women were excluded: Not available work-up (n= 48); Failed AI processing due to missing PACS image transfer or women with breast implant (n= 387); Women receiving only a single projection mammogram as follow-up studies of probably benign lesions (n= 16); studies of inadequate technical quality (N= 104). Image processing errors (due to breast implants, PACS transmission errors, poor technical quality, etc.) were pre-specified as exclusion criteria. Patients without available work-up were excluded from the study as their outcomes could not be determined.
Replication	The statistical analysis was cross checked internally to ensure correctness.
Randomization	This trial has a paired design, where each mammogram (either DM or DBT, depending on resource availability) has two reading strategies: the standard of care (double reading without AI support) and the AI-based intervention strategy (double reading with AI support if AI risk is higher than a predetermined threshold, alternatively, the exam will be automatically considered as normal). Thus randomization was not required.
Blinding	Readers were randomly assigned to each reading and blinded to other reading outcomes. Radiologists were aware of the strategy they were working on, given whether or not they had the support of AI.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.
Validation	Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.
Authentication	Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.
Mycoplasma contamination	Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.
Commonly misidentified lines (See ICLAC register)	Name any commonly misidentified cell lines used in the study and provide a rationale for their use.

Palaeontology and Archaeology

Specimen provenance	Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.
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Dating methods

they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.

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Ethics oversight

Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

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Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals

For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.

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Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.

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Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.

Field-collected samples

For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.

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Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

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Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration

NCT04949776

Study protocol

The Study protocol: Artificial intelligence in breast cancer screening program in Córdoba (AITIC), is publicly accessible via Zenodo at <https://doi.org/10.5281/zenodo.17625633>

Data collection

Data collected from Córdoba Breast Cancer Screening Unit, Córdoba, Spain, from March 15th 2022 to January 11th 2024.

Outcomes

The primary outcomes were workload, CDR and RR. Workload was computed as the absolute number of radiologists readings on each strategy. A cancer detected during the diagnostic work-up after a recall and confirmed by pathology was considered the gold standard. CDR was estimated as the number of cancers detected per 1000 screening examinations for each strategy. Cases that were not recalled, or were recalled without demonstrating malignancy on complementary imaging tests or biopsies, were considered negative. A minimum follow-up of 180 days after screening mammography was performed. Recall was defined as the screening examinations resulting in a call back for further assessment due to abnormal imaging findings. The recall rate was estimated as the number of women recalled per 100 screening examinations for each strategy. For each case, a strategy was positive if one or both readings included referral. Workload, CCR and RR were calculated globally and for DM and for DBT separately. As secondary objectives, PPV of recalls and FPR were assessed. The positive predictive value of recalls was calculated as the number of cancers diagnosed per 100 women recalled for each strategy. The false positive value was calculated as the number of women recalled who were not finally diagnosed with cancer for each strategy, per 100 negative screening examinations.

Dual use research of concern

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Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

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|-------------------------------------|---|
| No | Yes |
| ☒ | ☐ Public health |
| ☒ | ☐ National security |
| ☒ | ☐ Crops and/or livestock |
| ☒ | ☐ Ecosystems |
| ☒ | ☐ Any other significant area |

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Does the work involve any of these experiments of concern:

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| ☒ | ☐ Enable evasion of diagnostic/detection modalities |
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Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

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Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.

ChIP-seq

Data deposition

- ☐ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links

May remain private before publication.

For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.

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Methodology

Replicates

Describe the experimental replicates, specifying number, type and replicate agreement.

Sequencing depth

Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.

Antibodies

Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.

Peak calling parameters

Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.

Data quality

Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.

Software

Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.

Flow Cytometry

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.

Instrument

Identify the instrument used for data collection, specifying make and model number.

Software

Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.

Cell population abundance

Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.

Gating strategy

Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.

- ☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type

Indicate task or resting state; event-related or block design.

Design specifications

Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.

Behavioral performance measures

State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).

Acquisition

Imaging type(s)

Specify: functional, structural, diffusion, perfusion.

Field strength

Specify in Tesla

Sequence & imaging parameters

Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.

Area of acquisition

State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.

Diffusion MRI

☐ Used☐ Not used

Preprocessing

Preprocessing software

Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).

Normalization

If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.

Normalization template	Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.
Noise and artifact removal	Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).
Volume censoring	Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.

Statistical modeling & inference

Model type and settings	Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).
Effect(s) tested	Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.
Specify type of analysis:	☐ Whole brain ☐ ROI-based ☐ Both
Statistic type for inference	Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.
(See Eklund et al. 2016)	
Correction	Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).

Models & analysis

n/a	Involved in the study
☐	☐ Functional and/or effective connectivity
☐	☐ Graph analysis
☐	☐ Multivariate modeling or predictive analysis
Functional and/or effective connectivity	Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).
Graph analysis	Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).
Multivariate modeling and predictive analysis	Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.